Microscopic & dispersive dielectrics

EXTENSION: microscopic material models and frequency-dependent response, beyond the core capacitor chapter.

From dipoles to permittivity

$$p = \alpha E_{\text{local}}; \quad E_{\text{local}} = E + P/(3\varepsilon_0)$$

$$\frac{K-1}{K+2} = \frac{N\alpha}{3\varepsilon_0}$$

This Lorentz / Clausius-Mossotti model assumes an isotropic local-field description.

$$\alpha_e = \frac{e^2}{m\omega_0^2}, \quad \alpha_{\text{orient}} = \frac{p_0^2}{3k_BT}$$

WHY / Electronic polarisation is charge-cloud distortion; orientational polarisation is partial alignment of permanent dipoles against thermal agitation.

Orientational response weakens as temperature rises at fixed number density.

Debye relaxation and losses

Use time dependence exp(i omega t) and K* = K' - i K double-prime.

$$K^* = K_\infty + \frac{K_s - K_\infty}{1 + i\omega\tau}$$

$$K'' = \frac{(K_s - K_\infty)\omega\tau}{1 + (\omega\tau)^2}$$

$$\tan\delta = K''/K'$$

WHY / Dipoles cannot follow a rapidly changing field instantly. Their lag dissipates energy and makes the apparent permittivity frequency-dependent.

K double-prime peaks at omega tau = 1; tan delta peaks instead at sqrt(Ks/Kinfinity).

Heat, leakage and memory

$$P_{\text{loss}}/\mathcal{V} = \omega\varepsilon_0 K' \tan\delta\, E_{\text{rms}}^2$$

$$\tan\delta \simeq \omega C R_s \simeq \frac{1}{\omega C R_p}$$

Small-loss series ESR and parallel leakage are approximate equivalent models at a chosen frequency.

WHY / A real capacitor stores energy and dissipates some. Dielectric absorption adds slow internal modes: a discharged capacitor can recover voltage.

Ferroelectrics show hysteresis and bias/temperature dependence. Piezoelectric strain S = d33 E; free thickness change is d33 V.

WATCH / Stored energy is not generally CV squared/2 when C is strongly nonlinear.

Gas breakdown and small gaps

$$V_b = f(pd) \quad \text{(Paschen law)}$$

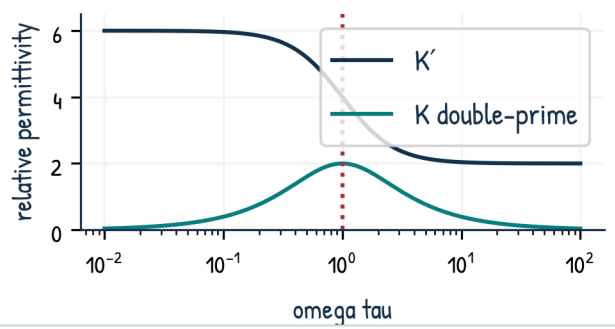
For classical gas discharge, breakdown depends on the product of gas pressure p and gap d. Air has a minimum near 330 V under the relevant discharge assumptions.

WHY / Avalanche growth competes with the probability of electron collisions. The pressure-gap product controls the number of opportunities for ionisation.

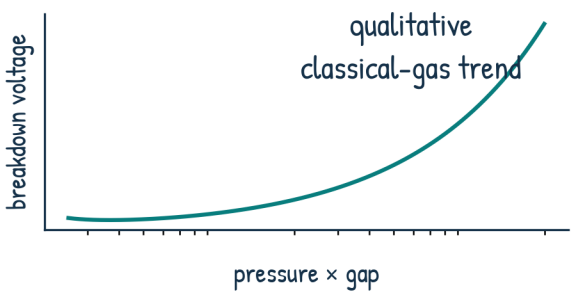
At very small gaps, field emission and tunnelling can invalidate a classical-gas model. Sharp features and voids concentrate local fields.

WATCH / The Paschen minimum is not an absolute voltage limit for every microscopic discharge mechanism.

Loss lags the response; K double-prime peaks at 1



Paschen minimum is not a universal microgap rule



REMEMBER / Distinguish intrinsic electrostatic storage from leakage, delayed polarisation and breakdown.

AC, dividers & timing circuits

EXTENSION: filters, compensated probes, threshold oscillators and slow-charging applications.

Impedance and the low-pass

$$i = C\frac{dV_C}{dt}; \quad Z_C = \frac{1}{i\omega C}$$

Sinusoidal current leads capacitor voltage by 90 degrees. For a series RC, output across C:

$$H(\omega) = \frac{1}{1 + i\omega RC}$$

$$|H| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$f_c = \frac{1}{2\pi RC}; \quad \phi = -\arctan(\omega RC)$$

WHY / At high frequency, C has small impedance and its output voltage is small. At cutoff: -3 dB and -45 degrees.

High-pass and compensated divider

Output across R in the same series RC:

$$H = \frac{i\omega RC}{1 + i\omega RC}$$

Compensated divider: upper arm R1 parallel C1, lower arm R2 parallel C2; output across lower arm.

$$\frac{V(0^+)}{V_s} = \frac{C_1}{C_1 + C_2}, \quad \frac{V(\infty)}{V_s} = \frac{R_2}{R_1 + R_2}$$

$$R_1C_1 = R_2C_2$$

WHY / Matching these time constants makes the fast capacitive division equal the slow resistive division, eliminating step-response distortion.

Relaxation timing

Charge from Vlo to Vhi toward Vs through R:

$$t_c = RC\ln\frac{V_s - V_{\text{lo}}}{V_s - V_{\text{hi}}}$$

Discharge from Vhi to Vlo toward zero through Rd:

$$t_d = R_dC\ln(V_{\text{hi}}/V_{\text{lo}})$$

For thresholds Vs/3 and 2Vs/3 and equal charge/discharge resistances: T = 2RC ln2.

WHY / A switching threshold converts a monotonic exponential into a repeated oscillation. The asymptotic voltage must lie beyond the threshold to reach it.

Ripple, gentle charging and noise

$$\Delta V \simeq I_{\text{load}}\Delta t/C$$

A reservoir capacitor supplies load charge between recharge events; choose C from the allowed ripple and longest recharge interval.

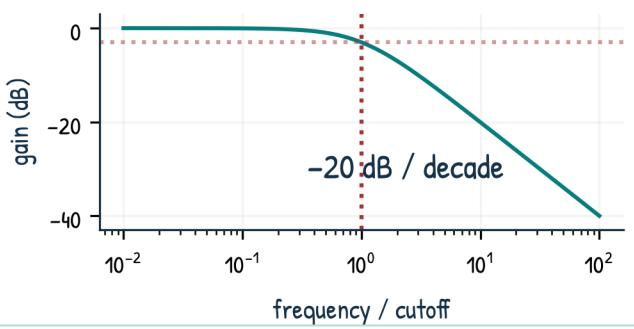
$$H_{\text{ramp}} \simeq RC^2V^2/T \quad (T \gg RC)$$

WHY / Slowly ramping the source keeps resistor current small, avoiding the unavoidable half-energy loss of a sudden voltage step.

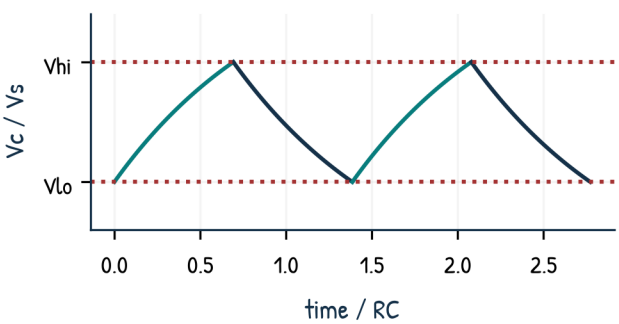
$$V_{\text{rms, noise}} = \sqrt{k_BT/C}$$

At 300 K and 1 pF: about 64 microvolt. Above self-resonance, lead inductance can dominate: a real capacitor is no longer capacitive.

RC low-pass across C: one pole, one time constant



Threshold switching repeats charge and discharge



REMEMBER / Always identify where the output voltage is measured before choosing a filter formula.

Coupled RC & specialised models

EXTENSION: coupled modes, measurement loading and exact wire-to-wire / wire-to-plane capacitances.

Several independent storage modes

$$\mathbf{C}\dot{\mathbf{V}} + \mathbf{G}\mathbf{V} = \mathbf{b}$$

Use independent node voltages after applying source and charge constraints. The matrices describe capacitive storage and resistive conduction.

$$V(t) = V_\infty + \sum_k A_k e^{-\lambda_k t}$$

WHY / Each independent relaxation mode has its own rate. A measured voltage can contain several exponentials, or hide a mode through symmetry.

For an invertible reduced capacitance matrix, rates are eigenvalues of C inverse G. Counting drawn capacitors alone is not sufficient.

An equal-R, equal-C example

Two nodes each have C to ground. An ideal source drives node 1 through R; another R joins nodes 1 and 2 (diagram below).

$$RC\dot{V}_1 = V_s - 2V_1 + V_2$$

$$RC\dot{V}_2 = V_1 - V_2$$

$$\lambda_\pm = \frac{3 \pm \sqrt{5}}{2RC}$$

WHY / The two node voltages can relax in different combinations. The eigenvalues of the conductance-storage system give two distinct time scales.

After a constant source step, both final voltages equal Vs. Initial voltages determine how strongly each mode appears.

Measuring can change the voltage

An isolated capacitor Cs at voltage V0 is connected to an initially uncharged meter input capacitance Cm. Neglect leakage during the reading.

$$V_{\text{read}} = V_0 \frac{C_s}{C_s + C_m}$$

WHY / The meter needs charge to raise its own input voltage. Charge sharing lowers the original source-capacitor voltage.

$$V_{\text{rms, noise}} = \sqrt{k_B T / C}$$

The thermal voltage scale is about 64 microvolt at 300 K and 1 pF. Small capacitance is sensitive, but more vulnerable to loading and noise.

Wires and the ground plane

Two equal long wires: radius a, centre spacing D > 2a, homogeneous surrounding dielectric.

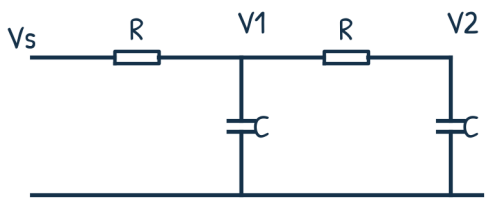
$$\frac{C}{L} = \frac{\pi\varepsilon}{\operatorname{arcosh}(D/2a)}$$

For D much larger than a, denominator becomes ln(D/a). One wire at centre height h > a above a grounded plane:

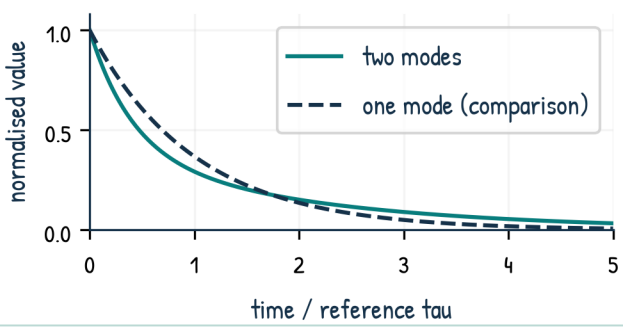
$$\frac{C}{L} = \frac{2\pi\varepsilon}{\operatorname{arcosh}(h/a)}$$

WHY / The ground plane is an equipotential midway between a wire and its opposite image. The wire-to-plane voltage is half the pair voltage.

Two storage nodes give two relaxation rates



Several storage modes need not share one time constant



REMEMBER / A single exponential is a model choice: identify the independent dynamic variables before assigning one time constant.

Matrices, reciprocity & images

Advanced tools simplify boundary conditions; they do not replace charge and energy bookkeeping.

Many conductors and reciprocity

$$Q_i = \sum_j C_{ij} V_j; \quad C_{ij} = C_{ji}$$

$$U = \tfrac{1}{2}\sum_i Q_i V_i = \tfrac{1}{2}\mathbf{V}^T\mathbf{C}\mathbf{V}$$

With a fixed reference: diagonal terms positive, off-diagonal terms nonpositive. If invertible, P = C inverse.

$$\sum_i Q_i V'_i = \sum_i Q'_i V_i$$

WHY / Reciprocity follows from symmetric C for the same geometry and linear dielectric.

Floating fixes Q; grounding fixes V.

Weighting potential and signal

Set the selected electrode to unit potential and every other electrode to zero, with the moving charge removed. Call the resulting potential phi_w.

$$\Delta Q_{\text{electrode}} = -q\Delta\phi_w$$

$$i_{\text{electrode}} = -q\vec{v}\cdot\nabla\phi_w$$

WHY / The weighting field measures how strongly each point couples to the selected electrode. It is not the actual driving field.

For parallel plates separated by d, phi_w = x/d when the selected electrode is at x = d. Current sign here means the rate of induced electrode charge.

Charge above a grounded plane

Replace a grounded plane by image -q at the mirror point. A real +q at height h feels:

$$|F| = \frac{q^2}{16\pi\varepsilon_0 h^2}$$

$$U = -\frac{q^2}{16\pi\varepsilon_0 h}$$

$$\sigma(\rho) = -\frac{qh}{2\pi(\rho^2 + h^2)^{3/2}}$$

WHY / The opposite image enforces V = 0 on the plane. The real force equals the image force, but induction energy is one-half q times image potential.

WATCH / Use separation 2h in Coulomb's force; the field is not that of an image only h away.

Grounded versus neutral sphere

Real q at distance d > a from a grounded sphere's centre. Place an image on the same radial line:

$$q' = -qa/d; \quad r' = a^2/d$$

$$|F| = \frac{q^2 ad}{4\pi\varepsilon_0(d^2 - a^2)^2}$$

WHY / The chosen image makes the entire spherical surface an equipotential at zero, not merely its nearest point.

For an isolated neutral sphere, add a central image +qa/d so total image charge is zero. Its force contribution must also be included.

WATCH / A grounded-sphere force formula is not automatically valid for a neutral floating sphere.

Real-image separation is 2h

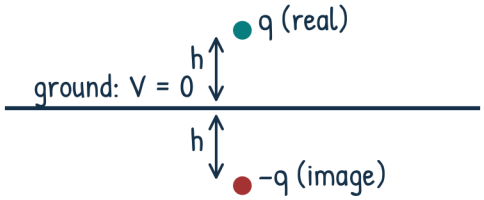
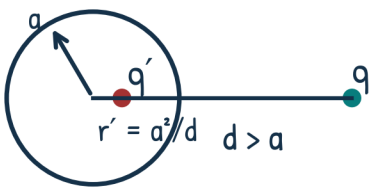


Image lies inside sphere; real charge is outside



REMEMBER / Image charges are mathematical replacements in the physical region—not extra real charges.

Shape, conformal maps & pores

Geometry controls both charge storage and where the electric field becomes dangerously large.

Spheroids: needle to disc

Semiaxis c along the symmetry axis, transverse semiaxis a. Prolate c > a; let f squared = c squared − a squared.

$$C = \frac{4\pi\varepsilon_0 f}{\ln[(c+f)/a]}$$

Oblate a > c: f squared = a squared − c squared; eta = arctan(f/c).

$$C = 4\pi\varepsilon_0 f/\eta$$

Limits: sphere C = 4 pi epsilon0 a; thin disc C = 8 epsilon0 a; long needle C approximately 4 pi epsilon0 c / ln(2c/a).

WHY / Tip curvature concentrates charge and local field. For an ellipsoidal inclusion the depolarising factor similarly controls its internal field.

Ellipsoidal inclusions and voids

$$E_{\text{in}} = \frac{E_0}{1 + L(\varepsilon_i/\varepsilon_m - 1)}$$

$$L_1 + L_2 + L_3 = 1$$

Here L is the depolarising factor along the applied field. Sphere: 1/3; axial needle: approaches 0; thin disc normal to field: approaches 1.

WHY / Polarisation charges on an inclusion's surface create an opposing field whose strength depends on shape.

An air disc-shaped crack normal to E in a K host sees about K E0. An axial needle-shaped void sees about E0. Thin cracks can therefore be much worse defects.

Wedges and corner fields

Two conducting faces at angles 0 and beta, radii r0 to R, held at 0 and V; beta in radians.

$$\phi = V\theta/\beta; \quad E_\theta = -V/(\beta r)$$

$$C' = \frac{\varepsilon}{\beta}\ln(R/r_0)$$

$$E_{\text{corner}} \propto r^{\,\pi/\beta - 1}$$

In the corner expression, beta is the vacuum opening angle between equipotential conductor faces. A knife edge has beta = 2 pi and E proportional to r to the minus one-half.

WHY / An analytic conformal map preserves 2D Laplace solutions; changing coordinates can turn a hard conductor shape into a simple strip.

Porous electrodes: delayed storage

Pore resistance per length r, capacitance per length c:

$$c\frac{\partial V}{\partial t} = \frac{1}{r}\frac{\partial^2 V}{\partial x^2}$$

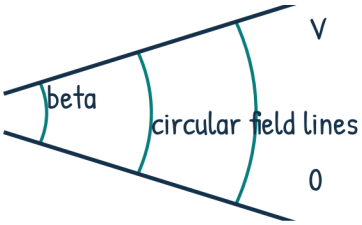
$$\ell_{\text{reached}} \sim \sqrt{t/(rc)}$$

$$Z_\infty = (1-i)\sqrt{r/(2\omega c)}$$

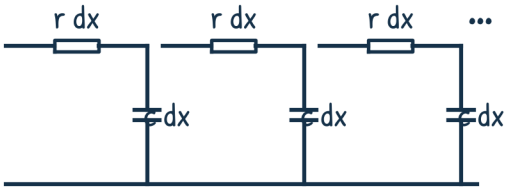
WHY / Charge must travel through resistance to reach deeper internal surface. Only part of the large geometric capacitance is accessible at short times.

Initially Q scales as square root of t. Full charging needs t of order rc L squared. A finite closed pore has low-frequency C = cL and ESR = rL/3.

Wedge: E falls as 1/r away from the apex



Distributed resistance delays access to deep pore walls



REMEMBER / A large total capacitance does not imply a small local field or rapid access to all stored charge.

Pull-in, oscillations & charged drops

EXTENSION: electromechanical instability and charged-drop shape stability, not a prerequisite for JEE capacitor problems.

Voltage-driven movable plate

Spring stiffness k, initial gap d0, closing displacement x; plate area A and homogeneous epsilon.

$$\Pi = \tfrac{1}{2}kx^2 - \frac{\varepsilon A V^2}{2(d_0 - x)}$$

$$kx = \frac{\varepsilon A V^2}{2(d_0 - x)^2}$$

WHY / Closing the gap strengthens voltage-driven attraction faster than the spring's restoring force. The stable and unstable equilibria can merge.

$$\Pi'' = k - \frac{\varepsilon A V^2}{(d_0 - x)^3} > 0$$

Pull-in and small oscillations

$$x_{\mathrm{pi}} = d_0/3$$

$$V_{\mathrm{pi}}^2 = \frac{8kd_0^3}{27\varepsilon A}$$

$$\omega^2 = \frac{1}{m}\left[k - \frac{\varepsilon A V^2}{(d_0 - x_{\mathrm{eq}})^3}\right]$$

WHY / Near pull-in, effective spring stiffness tends to zero, so the small-oscillation frequency softens to zero.

Use this formula only about a stable equilibrium and for small displacement. The one-third stroke limit assumes ideal parallel plates and a linear spring.

What changes at fixed charge?

$$U_Q = \frac{Q^2(d_0 - x)}{2\varepsilon A}$$

$$F = \frac{Q^2}{2\varepsilon A}; \quad x_{\mathrm{eq}} = \frac{Q^2}{2k\varepsilon A}$$

WHY / The ideal fixed-Q electric force is independent of gap. It shifts the spring equilibrium but does not soften its stiffness.

$$\omega = \sqrt{k/m}$$

There is no voltage-type fold before contact in this ideal model. Contact, fringing or charge leakage can invalidate it.

WATCH / Do not carry the voltage-driven pull-in result into a charge-controlled problem.

Rayleigh limit of a liquid drop

A conducting drop of radius R and surface tension gamma in vacuum has electric surface pressure:

$$p_e = \frac{Q^2}{32\pi^2\varepsilon_0 R^4}$$

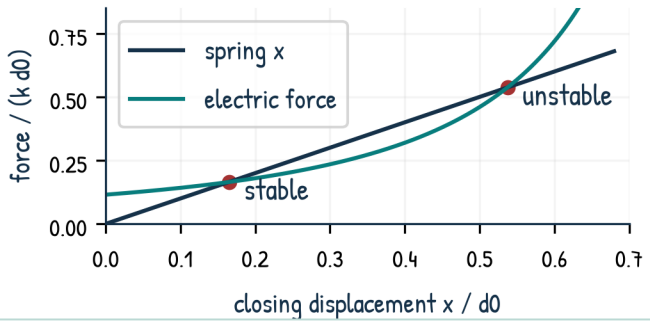
$$Q_R^2 = 64\pi^2\varepsilon_0\gamma R^3$$

$$V_R = 2\sqrt{\gamma R/\varepsilon_0}$$

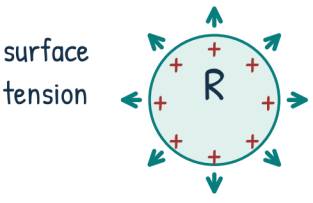
WHY / Above the Rayleigh limit, a shape deformation can lower total energy: surface tension can no longer stabilise the charged sphere.

This is a shape-instability result, not simply radial expansion at fixed volume. Real drops may also be limited by gas breakdown or charge emission.

Fixed V: lower-x crossing stable; higher-x unstable



Electric stress destabilises; surface tension stabilises



REMEMBER / A force-balance solution is not enough: test stability under the actual electrical constraint.